VU TRUONG VAN

EMBEDDED SOFTWARE ENGINEER

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OBJECTIVE

- Embedded Software Engineer with over 3 years of experience in bare-metal, RTOS, and event-driven programming for industrial embedded systems, including IoT devices, EV chargers, and AGVs.
- Experienced in firmware architecture design, firmware development, technical documentation, debugging, and software testing..
- Skilled in leading firmware development, mentoring engineers, conducting code reviews, and collaborating with cross-functional teams to deliver high-quality embedded solutions.

EDUCATION

Danang University of Science and Technology (DUT), Da Nang, Vietnam 2019 - 2023

Engineering in Mechatronics Engineering

SKILLS

Programming Languages	C, C++, Python, Bash
Embedded Software	Bare-metal, FreeRTOS, Event-driven Programming, Firmware Architecture, Driver Development.
MCUs	STM32, ESP32, Renesas, Pic, Arduino.
Communication Protocols	CAN, UART, SPI, I2C, RS-485, BLE, Wi-Fi, MQTT, HTTP, NMEA 2000.
Power Electronics	EV Charger, PFC Boost, LLC Resonant Converter, Battery Charging (CC/CV)
IoT & Connectivity	GNSS (AssistNow Online/Offline, CloudLocate, CellLocate), Cellular Communication
Unit Test	Ceedling, Unity
Debugging	JTAG, SWD, OpenOCD, Logic Analyzer, Oscilloscope, GDB
Tools	Vector CANoe, CANalyzer, Kvaser
Build & Dev Tools	CMake, Makefile, Docker

WORK EXPERIENCE

ASTI R&D VietNam Corporation 2025 - present

Technical Team Lead

- Led a firmware team of 2–3 engineers.
- Designed embedded software architecture.
- Analyzed customer requirements and defined technical solutions.
- Conducted code reviews and mentored engineers.
- Planned development tasks and coordinated project execution.
- Develop firmware for industrial applications such as: EV, IOT, Charger, AGV.
- Validate and testing performance for product.

ASTI R&D VietNam Corporation 2023 - 2025

Embedded Software Engineer

- Developed embedded firmware using C/C++ for industrial applications.
- Designed and implemented bare-metal,RTOS and event drivent programming.
- Implemented firmware update and data logging features for release products.
- Performed debugging, validation, and system testing.
- Developed Python tools for debugging and evaluation.
- Optimized embedded software performance and reliability.

Tan Long F&B Da Nang 04/2023 - 09/2023

Internship

- Designed and implemented the full pipeline from scratch: data analysis, data engineering workflows, computation, system design, and software and hardware development.

KEY - PROJECTS

EV - Charger Ac/Dc 800W 05/2026 - Present

Technical lead

- Analyze requirements and create software architecture.
- Design and implement charging control profiles CC/CV/CP.
- Develop open-loop and closed-loop control strategies for LLC power stages.
- Design, implement, and tune digital voltage and current PID control loops.
- Create and execute unit, integration, and system tests to verify software functionality and reliability.
- Validate charger performance, evaluate output power and optimize system efficiency.
- Collaborate with hardware team to troubleshoot system issues and support product release.
- Developed flash memory management for calibration data storage and fault/event logging.
- Developed low-level firmware drivers for Renesas MCU peripherals including ADC, PWM, timers, CAN, UART, and flash memory.
- Designed signal processing algorithms including EMA filtering, RMS estimation, and sensor calibration for stable and accurate measurements.

EV - Control Unit 11/2025 - 4/2026

Technical lead

- Designed RTOS-base firmware architecture, vehicle state machine and data flow.
- Developed vehicle features including RFID authentication, lighting system, horn, turn signals, buzzer, and other vehicle I/O functions.
- Designed and implemented CAN communication with Motor Inverter and Battery Management System (BMS).
- Utilized CANoe and CAN Analyzer for simulation, validation, and debugging of CAN communication and vehicle functionalities prior to full system integration.
- Developed telematics features using Cellular/GNSS module for remote monitoring, remote control, and GNSS tracking.
- Designed and developed LCD monitor using LVGL
- Designed and implemented firmware update mechanisms for vehicle ECUs.
- Performed module-level, integration-level, and system-level validation and testing.
- Led vehicle-level verification, real-world road testing, issue analysis, and validation for production release.

MPP Charger 08/2025 - 10/2025

Embedded Software Engineer

- Analyzed existing firmware architecture and source code of an established battery charger platform to understand system behavior and software design.
- Implemented and integrated new charging features for another battery types.
- Performed debugging, validation, and regression testing to ensure software reliability and product quality.
- Collaborated with hardware and system teams to verify new features and support product updates

MPP Charger System 02/2025 - 07/2025

Embedded Software Engineer

- Developedfirmware for ESP32-based main controller in a hybrid MPP charger system supporting AC, solar, and battery operating modes.
- Designed system state machines and charging workflows for power management and mode transitions.
- Implemented communication interfaces with battery via RS485 and power converters via UART.
- Developed protection and fault handling mechanisms including over-current, over-voltage, and system fault management.
- Implemented event and fault logging to support diagnostics and product release validation.
- Create multiple simulation and monitoring tools using Python for debugging and evaluation.

AGV - Automated Guided Vehicle -

Embedded Software Engineer

- Tune and optimize line following algorithm for factory environment using PID + Fuzzy
- Calibrate sensors to improve path detection accuracy

CERTIFICATIONS

• Level-Up C	08-2025
• C Programming for Embedded Applications	07-2025
• Test-Driven Development in C++	07-2025
• Mastering Micro-Controller and Embedded Driver Development	03-2025

HONORS & AWARDS

Software team's best employee of the year - ASTI R&D VIETNAM	12-2025
Software team's best employee of the year - ASTI R&D VIETNAM	12-2024

INTERESTS

Football, swimming, exploring new technologies, and finding ways to optimize engineering workflows and productivity.